…… Massive Crowd Management in Urban Pedestrian Zones

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***Abstract*—This paper presents a bi-level optimization framework for crowd management in large pedestrian zones, integrating anisotropic geometric guidance with discrete directional constraints. The lower level employs a Hughes-type macroscopic crowd model enhanced with metric tensor fields for channel guidance and constrained Hamilton-Jacobi-Bellman (HJB) equations for one-way channel control. The upper level optimizes channel direction configurations and guidance intensities using a Structure-Aware Hybrid Block Optimization (SA-HBO) algorithm. Three raw physical metrics are introduced for efficiency, safety, and load balancing, and are further converted into dimensionless indicators for cross-scheme comparison and scalarization. The proposed method exploits the local response characteristics of Bellman potential fields and employs surrogate pre-screening with Bellman warm-start strategies to reduce computational cost. Numerical experiments demonstrate the effectiveness of the proposed approach.**

***Index Terms*—Crowd dynamics, macroscopic modeling, channel optimization, Hamilton-Jacobi-Bellman equation, bi-level optimization.**

I. Introduction

LARGE-SCALE crowd gathering risks not only threaten individual safety but may also escalate into stampede incidents that disrupt social order and public security; the effectiveness of crowd management directly tests urban governance capacity and the modernization level of social safety governance systems [1], [2]. With the increasing frequency of high-density pedestrian activities such as holiday tourism, urban public events, commercial district promotions, New Year celebrations, and major sports events, core urban areas must accommodate pedestrian demands far exceeding daily levels within short time windows, posing enormous challenges for on-site organization, passage management, risk early warning, and emergency dispatch [3], [4], [5]. Against this backdrop, scientific modeling, numerical simulation, and optimization methods for formulating and evaluating crowd control strategies—improving the efficiency of large-scale event organization and reducing local congestion risks—have become critical scientific problems demanding urgent breakthroughs in social safety and urban governance [6], [7].

Crowds in large-scale events constitute typical complex systems. Numerous individuals with heterogeneous desired speeds, destinations, and route preferences interact through multi-scale, nonlinear mechanisms in confined spaces, generating spatiotemporal patterns at the collective level [8], [4]. Macroscopically, these patterns manifest as destination-oriented flow, passage convergence, bottleneck queuing, exit competition, and directional responses to control regulations [9], [10]; microscopically, individuals are influenced by surrounding density, visible paths, obstacle boundaries, and guidance facilities [11]. For open scenic areas, crowd behavior additionally exhibits pronounced phase dependence: visitors may first move toward the viewing platform, then tour along the main viewing direction, and subsequently choose different exit passages based on preferences [12]. These characteristics of multi-stage, multi-route coexistence with shared congestion mean that traditional static capacity analysis or single shortest-path models cannot adequately support refined management decisions.

Among various urban crowd gathering scenarios, open scenic areas, waterfront viewing platforms, and historic commercial districts are highly representative. Unlike enclosed venues such as stadiums and subway stations with well-defined boundaries [13], [14], such areas consist of open spaces, main tourist corridors, multiple lateral connecting passages, and return flow lines; visitors typically undergo a continuous behavioral chain of "entry–touring along the main corridor–choosing passages to leave–returning" [15], with crowd movement governed jointly by spatial geometry, control regulations, and historical behavioral preferences [16], [15]. Taking the Shanghai Nanjing East Road–Bund area as a concrete example, visitors approach the Bund from Nanjing East Road, ascend to the viewing platform via multiple stepped passages, tour along the waterfront, then exit through several southern passages and return along lower streets. This scenario can be abstracted as the typical "waterfront platform–multi-stepped passage" crowd flow scenario. In such scenarios, management difficulties manifest in three aspects: first, multiple passages simultaneously serve entry, exit, and diversion functions, and local direction settings can alter overall flow organization; second, distinct phase-dependent behavioral transitions exist between the main tourist corridor and lateral passages, making single origin–destination models insufficient; third, local congestion and global route choice form mutual feedback loops, where control changes in one passage may trigger load redistribution in adjacent areas. Fig. 1 shows the large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

Existing large-scale event security management practices typically integrate system dynamics, system coupling, and system safety theory, combined with high-risk zone delineation and historical crowd behavior patterns, to formulate a series of control measures [17], [18]. These measures include deploying security personnel at critical locations, erecting temporary barriers for isolation and diversion, implementing one-way passage on key channels, and applying batch release in core areas [19], [20]. For instance, during major holidays, the Shanghai Bund area adopts batch release, passage direction regulation, south-entry north-exit routing, and reduction of head-on conflicts [21]; similarly, Nanluoguxiang main street in Beijing implements one-way passage measures (south entry,

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Fig. 1. Large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

north exit) during peak periods [22], consistent in control mechanism with the Shanghai Bund approach.

However, current management measures for urban high-pedestrian-flow areas still rely considerably on manual experience and static contingency plans, with deficiencies remaining in scientific evaluation and refined optimization [23]. First, many control schemes are formulated based on historical experience, lacking quantitative characterization of the impact of different passage direction configurations, geometric guidance intensities, and phase-dependent behavioral preferences. Second, existing methods tend to focus on single-point statistics or local density monitoring, making it difficult to describe the dynamic coupling among "passage direction rules–local optimal directions–collective density evolution–exit load distribution" [24]. Third, in open scenic area scenarios with multiple entrances, passages, and stages, neglecting visitor route preferences and phase transition behaviors can easily lead to misjudgment of passage loads and high-risk area locations. Finally, many existing simulation or optimization methods treat the crowd model as a black-box evaluator, failing to exploit the structural information among passage geometry, one-way rules, and multi-route subpopulations [25], resulting in insufficient efficiency in control strategy search [26].

To address the above issues, this paper proposes a macroscopic crowd modeling and control optimization method incorporating one-way passage constraints, geometric guidance, and multi-stage route behavior, targeting the "waterfront platform–multi-stepped passage" open scenic area scenario. At the modeling level, this paper constructs a Bellman–conservation law coupled crowd continuum model: different stage–route subpopulations possess independent potential functions with optimal directions computed via the discrete Bellman equation, while sharing the speed–density relationship determined by total density; one-way passage rules are formulated as constraints on the local admissible direction set, and an anisotropic metric tensor characterizes the cost differences between axial and lateral movement, simulating pedestrian pre-alignment and streamline reorganization near passages. At the optimization level, this paper distinguishes "controllable management variables" (passage direction configuration and geometric guidance intensity) from "exogenous behavioral parameters" (visitor route diversion preferences identified from historical data). Based on this boundary, total travel time, high-density exposure time, and channel cumulative flow variance are adopted as efficiency, safety, and load balancing indicators, forming a scalarized optimization problem under fixed weights. A simplified instance of the Structure-Aware Hybrid Block Optimization framework (SA-HBO) is designed to solve this problem, exploiting the discrete neighborhood structure of passage direction variables, the continuous nature of geometric guidance variables, and the response characteristics of Bellman potential fields to local control changes, improving candidate strategy evaluation efficiency through surrogate pre-screening, full simulation confirmation, and Bellman warm-start procedures.

The main contributions of this paper are as follows.

1) A "waterfront platform–multi-stepped passage" crowd flow modeling framework for open scenic areas is proposed, representing visitor behavior as a multi-stage, multi-route continuum flow process, where different subpopulations possess independent potential functions while sharing collective congestion feedback.
2) A Bellman–conservation law coupled model integrating one-way passage constraints and geometric guidance is constructed, enabling control rules to directly affect local optimal direction selection, with an anisotropic metric tensor simulating pre-alignment, smooth convergence, and streamline reorganization near passage entrances.
3) An evaluation framework distinguishing mechanism verification from strategy optimization is established, categorizing indicators into behavioral mechanism indicators (direction consistency, approach angle distribution, channel capture domain) and management optimization indicators (total travel time, high-density exposure time, channel flow variance).
4) A structure-aware optimization framework for controllable passage configuration is designed, jointly optimizing passage direction configuration and geometric guidance intensity under fixed historically identified preference parameters through discrete neighborhood search, continuous local search, and surrogate pre-screening with acceptance/rejection mechanisms.

II. Related Work

A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [27], [28].

In control-theoretic approaches, Zhong et al. [29] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [30]. For one-dimensional crowd dynamics, Wadoo and Kachroo [31]

employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [32] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [33] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [23], [34]. Yang et al. [35], [36] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [37] proposed dynamic exit-choice recommendations using multinomial Logit models. However, their approach relies on discrete speed instructions (e.g., "stop," "walk," "run") transmitted via electronic wristbands [38], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [39] formulated fence layout as a subset selection problem, while subsequent work [26] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [40] for real-time entrance flow regulation. Differential Evolution (DE) [41] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [42].

B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency and analytical tractability compared to microscopic approaches. Henderson [43] first applied fluid dynamics to pedestrian traffic in the early 1970s, comparing crowd motion to the behavior of gas or fluid particles.

Hughes [44] pioneered the first-order continuum theory for pedestrian flow, introducing the classical model in which pedestrian walking speed is governed by local density through a fundamental diagram, while movement direction follows the steepest-descent path of a potential function representing the remaining travel time to exits. This framework elegantly captures the interplay between congestion and route choice. Ling et al. [45] subsequently generalized the Hughes formulation and developed an efficient numerical solution combining a weighted essentially non-oscillatory (WENO) scheme for the conservation law with a fast sweeping method for the eikonal equation.

Several extensions have enriched the descriptive capability of macroscopic models. Nonlocal flux models were proposed by Colombo et al. [46], replacing pointwise density with a neighborhood-averaged quantity in the velocity function to account for anticipatory behavior. This direction was further extended to incorporate anisotropic perception fields and boundary effects from walls and obstacles, with high-resolution numerical schemes and well-posedness results suitable for realistic architectural configurations [47]. To enforce the hard incompressibility constraint—that pedestrian density cannot exceed a physically admissible maximum—Maury et al. [48] introduced the projection model, which formulates the actual velocity as a least-squares projection of the desired velocity onto the set of feasible (non-overlapping) configurations. This framework was later recast as a Wasserstein gradient flow with density constraints [49], using the Jordan–Kinderlehrer–Otto (JKO) scheme [50] to discretize the motion while guaranteeing that density respects an upper bound, making it well-suited for high-density congestion scenarios.

To overcome non-physical behaviors such as supersonic information propagation that may arise in first-order models, Aw and Rascle [51] introduced a second-order model based on a generalized momentum conservation law that preserves anisotropy. The Aw–Rascle (AR) framework was subsequently adapted to pedestrian flows [52], where the congestion pressure term successfully reproduces characteristic phenomena including stop-and-go waves and spontaneous lane formation.

Beyond the classical Hughes-type framework, extensions have incorporated control and guidance mechanisms relevant to crowd management. Anisotropic formulations using metric tensors have been developed to represent channel geometries and directional preferences [53], [54], allowing for continuous geometric guidance without explicit prohibition. Constrained Hamilton-Jacobi-Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors, with the Bellman optimality principle providing a natural framework for unifying anisotropic geometric guidance with discrete directional constraints.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The bi-level optimization of discrete channel direction configurations combined with continuous guidance intensities remains underexplored, particularly for large-scale urban pedestrian zones with multiple behavioral stages and heterogeneous route choices.

III. Macroscopic Crowd Dynamics Model

A. Hughes Model Foundation

We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walkway. Let $\rho(x,t)$ denote the crowd density at position x and time t, $\phi(x,t)$ the minimal remaining time to reach an exit, and $\mathbf{v}(x,t)$ the mean velocity field. The continuity equation governing mass conservation reads:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{1}$$

In the classical Hughes model [44], pedestrians move in the direction of steepest descent of the potential function:

$$\mathbf{v} = f(\rho)\frac{-\nabla \phi}{|\nabla \phi|}, \tag{2}$$

where $f(\rho)$ is the speed-density relationship. We adopt the Greenshields form:

$$f(\rho) = v_{\max}\left(1 - \frac{\rho}{\rho_{\max}}\right), \tag{3}$$

with $v_{\max}$ the free-flow speed and $\rho_{\max}$ the jam density.

The potential function satisfies the eikonal equation:

$$|\nabla\phi| = \frac{1}{f(\rho)}. \tag{4}$$

While this model captures basic congestion effects, it assumes isotropic movement and cannot represent channel geometries or directional controls. We address these limitations in the following subsections.

B. Channel Geometric Guidance via Metric Tensor

To represent anisotropic channel geometries such as corridors and barriers, we introduce a spatially varying symmetric positive-definite metric tensor $\mathbf{M}(x)$ that generalizes the classical eikonal equation to anisotropic form [53]:

$$\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi} = \frac{1}{f(\rho)}. \tag{5}$$

Under this metric, the optimal direction is determined by the tensor-induced steepest descent:

$$\mathbf{v} = f(\rho)\frac{-\mathbf{M}(x)\nabla\phi}{\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi}}. \tag{6}$$

For a channel region Ω_c with unit tangent $\tau(x)$ and unit normal $n(x)$, we construct the metric tensor as:

$$\mathbf{M}(x) = \alpha(x)\tau\tau^\top + \beta(x)nn^\top, \quad x \in \Omega_c, \tag{7}$$

where $\alpha(x)$ and $\beta(x)$ control the traversal weights in tangential and normal directions, respectively. Setting $\alpha(x) \gg \beta(x)$ encourages movement along the channel axis while suppressing lateral crossing, achieving continuous geometric guidance without explicit prohibition. Outside channels, $\mathbf{M}(x) = \mathbf{I}$ recovers the isotropic case.

Remark: The metric tensor approach represents directional preferences rather than hard constraints. As long as costs remain finite, the model allows individuals to deviate from preferred directions when facing congestion or detour requirements. This motivates the introduction of constrained HJB framework for strict directional controls.

C. Channel Control via Constrained HJB

Many crowd management measures directly restrict available movement directions rather than merely adjusting traversal costs. Examples include one-way corridors and temporary closures. Such rules are naturally modeled via constrained optimal control.

Let $U(x)$ denote the set of admissible unit directions at position x. The velocity satisfies:

$$\mathbf{v} = f(\rho)\mathbf{u}, \quad \mathbf{u}(x) \in U(x), \quad |\mathbf{u}| = 1. \tag{8}$$

By the Bellman optimality principle, the potential function satisfies the constrained HJB equation:

$$\max_{\mathbf{u}\in U(x)} \{-f(\rho)\mathbf{u}\cdot\nabla\phi\} = 1. \tag{9}$$

When $U(x)$ is the full unit circle, (9) reduces to the classical Hughes model. For restricted control sets, various channel rules can be expressed naturally. For a strict one-way channel with allowed direction $\tau(x)$:

$$U(x) = \{\tau(x)\}, \quad x \in \Omega_c, \tag{10}$$

and the HJB equation becomes:

$$-f(\rho)\tau(x)\cdot\nabla\phi = 1, \tag{11}$$

encoding one-way movement as a signed directional constraint.

D. Unified Discrete Bellman Formulation

To unify anisotropic geometry with discrete constraints, we employ a discrete Bellman formulation. For grid point x and admissible direction $\mathbf{u} \in U(x)$, define the step cost:

$$\text{Cost}(x, \mathbf{u}) = \frac{\Delta x}{f(\rho(x))}\frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}(x)\mathbf{u}}}, \tag{12}$$

The potential satisfies:

$$\phi(x) = \min_{\mathbf{u}\in U(x)} \left[\phi(x + \Delta x\,\mathbf{u}) + \text{Cost}(x, \mathbf{u})\right]. \tag{13}$$

This formulation couples metric tensor $\mathbf{M}(x)$ for geometric guidance with control set $U(x)$ for directional constraints within a unified Bellman framework. The optimal direction is:

$$\mathbf{u}^*(x) = \arg\min_{\mathbf{u}\in U(x)} \left[\phi(x + \Delta x\,\mathbf{u}) + \text{Cost}(x, \mathbf{u})\right], \tag{14}$$

with velocity $\mathbf{v}(x) = f(\rho)\mathbf{u}^*(x)$.

E. Multi-Stage Multi-Route Model

Real-world scenarios often involve complex itineraries with multiple stages and route choices. We extend the density field to include stage and route indices, with fixed-probability splitting at decision boundaries.

1) Stage-Route Decomposition: Let the complete behavior chain consist of S stages, where stage s has R_s alternative routes. The total density decomposes as:

$$\rho(x, t) = \sum_{s=1}^{S}\sum_{r=1}^{R_s} \rho_{s,r}(x, t), \tag{15}$$

where $\rho_{s,r}(x, t)$ is the density of sub-population in stage s following route r.

All sub-populations share the same speed function $f(\rho)$ determined by total density, ensuring collective congestion effects. Sub-populations differ in their potential functions, optimal directions, and exit targets.

2) Sub-Population Potentials and Velocities: For each sub-population (s, r), define admissible direction set $U_{s,r}(x)$, metric tensor $\mathbf{M}_{s,r}(x)$, and target regions. The potential $\phi_{s,r}(x)$ satisfies:

$$\phi_{s,r}(x) = \min_{\mathbf{u}\in U_{s,r}(x)} \Big[\phi_{s,r}(x + \Delta x\, \mathbf{u}) + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x)\mathbf{u}}} \Big]. \tag{16}$$

The optimal direction is:

$$\mathbf{u}^*_{s,r}(x) = \arg\min_{\mathbf{u}\in U_{s,r}(x)} \Big[\phi_{s,r}(x + \Delta x\, \mathbf{u}) + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x)\mathbf{u}}} \Big], \tag{17}$$

with velocity $\mathbf{v}_{s,r}(x) = f(\rho)\, \mathbf{u}^*_{s,r}(x)$.

3) Stage Transitions and Fixed-Probability Splitting: Sub-populations couple through stage transitions. Let $G_{s,r} \subset \Omega$ be the completion goal region for stage (s, r) with indicator $\chi_{G_{s,r}}(x)$. Upon reaching this region, pedestrians switch to next-stage routes with fixed probabilities.

Let $p_{(s,r)\to q}$ be the fixed probability of switching from (s, r) to $(s + 1, q)$, satisfying:

$$p_{(s,r)\to q} \geq 0, \quad \sum_{q=1}^{R_{s+1}} p_{(s,r)\to q} = 1. \tag{18}$$

The transition term is:

$$Q_{(s,r)\to(s+1,q)}(x, t) = p_{(s,r)\to q}\, \kappa_{s,r}\, \chi_{G_{s,r}}(x)\, \rho_{s,r}(x, t), \tag{19}$$

where $\kappa_{s,r} > 0$ is the stage switching rate.

The density evolution for each sub-population includes advection and transition source terms:

$$\frac{\partial \rho_{s,r}}{\partial t} + \nabla \cdot (\rho_{s,r}\mathbf{v}_{s,r}) = Q^{\text{in}}_{s,r} - Q^{\text{out}}_{s,r}, \tag{20}$$

where $Q^{\text{in}}_{s,r}$ is inflow from the previous stage and $Q^{\text{out}}_{s,r}$ is outflow to the next stage.

IV. Numerical Solution

We discretize the domain using a uniform two-dimensional grid, with the control direction set represented as an 8-neighborhood basis with masking for regional restrictions.

Unified Bellman Solver. Rather than separate eikonal and HJB solvers, we use a unified discrete Bellman shortest-path update. For candidate direction $\mathbf{u}$, precompute geometric step factors from the local tensor, then combine with the speed function to form the complete step cost. A priority queue propagates potential values outward from exits until convergence. When the control set is unrestricted, this procedure is equivalent to monotone solution of the anisotropic eikonal equation; when controls are restricted, it naturally reduces to constrained HJB solution, covering all required scenarios within a unified framework.

Density Update. After obtaining converged potentials, recover velocity fields using (17), substitute into the continuity equation, and advance density explicitly using an upwind flux scheme:

$$\rho^{n+1} = \rho^n - \Delta t\, \nabla \cdot (\rho^n \mathbf{v}^n). \tag{21}$$

The time step satisfies the CFL stability condition.

Evaluation Oracle. For any given control configuration $z = (s, \eta)$, the simulation workflow constitutes a lower-level evaluation operator:

$$(\rho, \phi, \mathbf{v}) = \mathcal{S}(z; \hat{p}), \tag{22}$$

where $\hat{p}$ represents fixed route preference parameters identified from historical data. This operator provides the foundation for subsequent control optimization.

V. Control Evaluation Metrics

We distinguish between behavior validation metrics (for verifying model mechanisms) and control evaluation metrics (for assessing policy impacts). This section focuses on the latter.

A. Efficiency Metric: Total Travel Time

For simulation horizon $[0, T]$, the total travel time metric is:

$$J_1(z \mid \hat{p}) = \int_0^T \int_\Omega \rho(x, t)\, dx\, dt. \tag{23}$$

This measures aggregate dwell time of the crowd, providing a stable basis for comparing different control strategies.

B. Safety Metric: High-Density Exposure Time

Let ρ_{safe} be the safe density threshold. The exposure time metric is:

$$J_2(z \mid \hat{p}) = \int_0^T \int_\Omega \mathbf{1}_{\rho(x,t)>\rho_{\text{safe}}}\, dx\, dt. \tag{24}$$

This captures cumulative space-time exposure to dangerous crowding conditions, reflecting management concerns about sustained congestion risks.

C. Balance Metric: Channel Flow Variance

Let $q_c(t)$ be the flux through channel c at time t, with cumulative flow $F_c = \int_0^T q_c(t)\, dt$. The load balancing metric is:

$$J_5(z \mid \hat{p}) = \text{Var}(F_1, \ldots, F_m). \tag{25}$$

Lower variance indicates more even distribution of traffic across channels, reducing risk concentration and improving facility utilization.

D. Dimensionless Normalization

Because J_1, J_2, and J_5 have different physical units and numerical scales, we further define normalized indicators for policy comparison and optimization. Let $M_{\text{tot}} = M_{\text{init}} + M_{\text{in}}$ denote the total evaluated crowd mass, let $|\Omega_w|$ be the walkable area, and let $F = \sum_{c=1}^m F_c$ be the total cumulative channel flow. We define

$$\tilde{J}_1(z \mid \hat{p}) = \frac{J_1(z \mid \hat{p})}{M_{\text{tot}} T}, \tag{26}$$

$$\tilde{J}_2(z \mid \hat{p}) = \frac{J_2(z \mid \hat{p})}{|\Omega_w| T}, \tag{27}$$

and

$$\tilde{J}_5(z \mid \hat{p}) = \frac{\mathrm{Var}(F_1, \ldots, F_m)}{F^2(m-1)/m^2} = \frac{m^2}{m-1}\mathrm{Var}(p_1, \ldots, p_m), \tag{28}$$

where $p_c = F_c/F$. The normalized metrics are dimensionless, with $\tilde{J}_2 \in [0,1]$ and $\tilde{J}_5 \in [0,1]$.

E. Scalarized Optimization Objective

Given weights $(\lambda_1, \lambda_2, \lambda_5)$, the scalarized objective is:

$$J(z \mid \hat{p}) = \lambda_1 \tilde{J}_1(z \mid \hat{p}) + \lambda_2 \tilde{J}_2(z \mid \hat{p}) + \lambda_5 \tilde{J}_5(z \mid \hat{p}), \tag{29}$$

with the optimization problem:

$$\min_{z \in \mathcal{Z}} J(z \mid \hat{p}). \tag{30}$$

Our algorithm solves this scalarized problem for fixed weights, rather than directly generating the full Pareto frontier. Approximate Pareto sets can be obtained by running with different weight combinations.

VI. Optimization Method

A. Control Variable Definition

We distinguish controllable variables from exogenous behavioral parameters:

Controllable variables:

1) *Channel direction configuration* $s = (s_1, \ldots, s_C)$, where $s_c \in \{-1, 0, +1\}$ indicates reverse one-way, bidirectional, or forward one-way for channel c.
2) *Geometric guidance intensity* $\eta = (\eta_1, \ldots, \eta_C)$, where $\eta_c = \alpha_c/\beta_c \geq 1$ is the anisotropy ratio.

Thus the optimization variable is $z = (s, \eta)$.

Exogenous parameters: Route preference parameters $\hat{p}$ are identified from historical data and fixed during optimization. Decision region locations are reserved for future extension.

B. Structure-Aware Hybrid Block Optimization

Treating this as a black-box optimization ignores three key structures: mixed discrete-continuous variables, local Bellman response characteristics, and the expensive multi-group simulation structure.

We propose Structure-Aware Hybrid Block Optimization (SA-HBO), with four core components:

1) Discrete block neighborhood search for channel directions;
2) Surrogate pre-screening to reduce full simulations;
3) Bellman warm-start with local re-solving for efficient evaluation;
4) Continuous block projected stochastic approximation for guidance intensities.

C. Discrete Block Update

Fixing continuous variables η^k, the discrete subproblem at iteration k is:

$$s^{k+1/2} \approx \arg\min_{s \in \mathcal{S}} J(s, \eta^k \mid \hat{p}). \tag{31}$$

Define the neighborhood of current configuration s^k as:

$$\mathcal{N}(s^k) = \left\{ s : \|s - s^k\|_0 \leq \nu \right\}, \tag{32}$$

where ν is the maximum number of simultaneously flipped channels (typically 1 or 2).

Surrogate pre-screening: To reduce full simulations, construct a local surrogate for each candidate $s' \in \mathcal{N}(s^k)$:

$$\widehat{J}(s', \eta^k) = \mu_1 \widehat{J}_1^{\text{local}} + \mu_2 \widehat{J}_2^{\text{local}} + \mu_3 \widehat{J}_3^{\text{flow}}, \tag{33}$$

where the local metrics approximate density changes, exposure changes, and flow redistribution. Select top K candidates by surrogate value for full PDE simulation.

D. Continuous Block Update

Fixing discrete variables $s^{k+1/2}$, solve for η:

$$\eta^{k+1} \approx \arg\min_{\eta \in \mathcal{H}} J(s^{k+1/2}, \eta \mid \hat{p}), \tag{34}$$

with feasible set $\mathcal{H} = \{\eta : \eta_c \geq 1\}$.

Since gradients are unavailable analytically, we use projected stochastic approximation. Generate Rademacher perturbation $\Delta_\eta^k \in \{-1, +1\}^C$, construct two-sided perturbations $\eta_\pm^k = \eta^k \pm c_k \Delta_\eta^k$, and project to feasible set. The stochastic gradient estimate is:

$$\widehat{g}_\eta^k = \frac{J(s^{k+1/2}, \tilde{\eta}_+^k \mid \hat{p}) - J(s^{k+1/2}, \tilde{\eta}_-^k \mid \hat{p})}{2c_k} \odot (\Delta_\eta^k)^{-1}. \tag{35}$$

The candidate update is:

$$\eta_{\text{cand}}^{k+1} = \Pi_{\mathcal{H}}\left(\eta^k - \alpha_k D_\eta^{-1} \widehat{g}_\eta^k\right), \tag{36}$$

with diagonal preconditioner D_η.

Acceptance/Rejection: Compare candidate objective J_{cand} with current J_{cur}. Accept if $J_{\text{cand}} \leq J_{\text{cur}} - \varepsilon_{\text{acc}}$; otherwise reject and optionally backtrack. Maintain best solution z_{best} throughout.

E. Bellman Warm-Start and Local Re-solving

Control changes typically affect only local regions. We identify the affected region Ω_{chg}^k where $U_{s,r}(x; \tilde{z}) \neq U_{s,r}(x; z^k)$ or $\mathbf{M}_{s,r}(x; \tilde{z}) \neq \mathbf{M}_{s,r}(x; z^k)$. Using the old potential as initialization, iterate only within an expanded neighborhood $\Omega_{\text{chg}}^{k,\delta}$ until convergence.

This accelerates candidate evaluation. Periodic global corrections prevent error accumulation.

F. Algorithm Summary

1) Initialize potentials and density, compute $J(z^0 \mid \hat{p})$, set $z_{\text{best}} = z^0$.
2) For each iteration:
   - *Discrete update*: Generate neighborhood, screen with surrogate, evaluate top candidates with Bellman warm-start, select $s^{k+1/2}$.
   - *Continuous update*: Generate perturbations, compute stochastic gradient, generate candidate, accept/reject to obtain η^{k+1}.
   - *State update*: Set $z^{k+1} = (s^{k+1/2}, \eta^{k+1})$, compute objective, update z_{best} if improved.
3) Return $z^\star = z_{\text{best}}$.

Stopping criteria include: small objective change, no discrete variable change for consecutive iterations, or maximum iteration/simulation/time limits.

VII. Numerical Experiments

This section presents numerical experiments to validate the model mechanisms, evaluate control strategies, and demonstrate the optimization framework. The experimental design follows a two-layer validation logic: first verifying whether the model mechanisms authentically alter local crowd behaviors, then comparing different channel control strategies in terms of efficiency, safety, and load balancing.

A. Experimental Setup

1) Three-Channel Scenario: The experiments employ a three-channel scenario as the primary testbed for mechanism validation and strategy comparison. The geometry consists of an initial crowd region on the left, an exit region on the right, and three passages (upper, middle, lower) through a central wall separating the two regions.

Numerical parameters are set as follows: grid resolution 96×72, spatial step $\Delta x = 0.5$, free-flow speed $v_{\max} = 1.5$, maximum density $\rho_{\max} = 5.0$, initial density $\rho_{\text{init}} = 2.2$, CFL coefficient 0.9, maximum simulation steps 600, and time horizon $T = 40$.

2) Evaluation Metrics: The experiments employ three primary raw metrics and their normalized counterparts:

- *Efficiency metric* J_1 and $\tilde{J}_1$: total travel time and its mass-time normalized form;
- *Safety metric* J_2 and $\tilde{J}_2$: high-density exposure time and its space-time normalized form;
- *Balance metric* J_5 and $\tilde{J}_5$: channel flow variance and its normalized imbalance index.

Behavioral indicators include channel capture domain ratios, flow distribution ratios, direction consistency indices, and approach angle distributions.

B. Mechanism Validation

The mechanism validation experiments address two fundamental questions: (1) Do geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local decision-making and flow patterns? (2) Does the model produce sufficiently sensitive and interpretable responses to configuration changes?

TABLE I
Bidirectional $U(x)$ Validation Results

Configuration	Westbound Share	Counter-flow Mass	Head-on Exposure
Bidirectional baseline	73.19%	152.15	1055.74
Middle channel one-way	0	0	0

TABLE II
Configuration Sensitivity Results

Config	Dominant Capture	Dominant Flow	Upper	Middle	Lower
middle	middle	middle	∼1/3	dominant	∼1/3
top	top	top	dominant	∼1/3	∼1/3
bottom	bottom	bottom	∼1/3	∼1/3	dominant

1) Effect of Geometric Guidance $M(x)$: To verify whether geometric guidance alters approach patterns and attraction domains, we compare the baseline configuration (neither M nor U) against the M-only configuration (anisotropic guidance applied to the middle channel).

Results: Under the baseline configuration, the three channels exhibit approximately balanced capture domains: upper 33.73%, middle 30.00%, lower 34.98%. Flow distributions are similarly balanced: upper 33.82%, middle 32.35%, lower 33.83%.

Under the M-only configuration, the middle channel capture domain expands significantly to 63.41%, and its flow share jumps to 99.32%. This demonstrates that geometric guidance $M(x)$ reshapes local attraction domains and fundamentally alters approach patterns.

2) Effect of Directional Constraints $U(x)$: To verify whether directional constraints alter local optimal directions and channel selection, we compare the baseline against the U-only configuration (middle channel restricted to eastward-only flow).

Results: Under the U-only configuration, the middle channel capture domain drops to 0, and its flow share drops to 0, with upper and lower channels splitting traffic approximately $50/50$. This confirms that $U(x)$ significantly rewrites feasible path choices.

To further verify the suppression of counter-flow and head-on conflicts, we conduct a bidirectional flow sub-experiment. Two configurations are tested: bidirectional baseline (westbound and eastbound crowds simultaneously) and bidirectional with middle channel one-way rule.

The results in Table I show that when the middle channel enforces a one-way rule, westbound crowds are completely excluded, and counter-flow mass time and head-on cell exposure are both driven to zero. This confirms that $U(x)$ not only changes path selection but effectively suppresses counter-flow and head-on conflicts under bidirectional demand.

3) Combined $U + M$ Effects and Configuration Sensitivity: To verify whether the combined $U + M$ model produces distinguishable responses to different channel configurations, we test three setups: middle-channel-controlled ($U + M$-middle), upper-channel-controlled ($U + M$-top), and lower-channel-controlled ($U + M$-bottom).

Table II demonstrates that the $U + M$ model exhibits stable and interpretable responses to configuration changes: when the middle channel is controlled, it becomes the dominant

capture and flow channel; when the upper or lower channel is controlled, the corresponding channel becomes dominant. This sensitivity validates the model as a foundation for subsequent strategy comparison and optimization.

4) Mechanism Validation Conclusions: The experimental results support four key conclusions:

1) Geometric guidance $M(x)$ significantly alters local approach patterns and channel attraction domains, enabling earlier alignment with channel axes and smoother convergence.
2) Directional constraints $U(x)$ reshape feasible paths and directional organization, effectively suppressing counter-flow and head-on conflicts under bidirectional demand.
3) The combined $U + M$ model produces configuration-sensitive responses that are both rule-compliant and behaviorally realistic.
4) Either U or M alone yields incomplete or behaviorally unrealistic responses; the full model is necessary for reliable optimization.

C. Control Strategy Comparison

This subsection compares different channel control strategies in terms of normalized efficiency ($\tilde{J}_1$), normalized safety ($\tilde{J}_2$), and normalized load balancing ($\tilde{J}_5$). The strategies under comparison include:

- Baseline (no control);
- Upper-channel-guided;
- Middle-channel-guided;
- Lower-channel-guided.

Detailed results and analysis will be presented in the full version of this paper.

D. Multi-Stage Flow Validation Experiments

This subsection validates the multi-stage behavior chain and probabilistic splitting mechanism. The experiments compare single-stage approximations against full multi-stage models to demonstrate how stage transitions and fixed-probability splitting alter density evolution and exit load distribution.

Detailed results and analysis will be presented in the full version of this paper.

E. Optimal Control Parameter Search

This subsection presents the first-version automated parameter search results. The search enumerates discrete strategy variables, performs coarse grid search on guidance intensity η, and searches over fixed splitting probability candidates to identify optimal control configurations under different objective weight combinations $(\lambda_1, \lambda_2, \lambda_5)$.

Detailed results and analysis will be presented in the full version of this paper.

VIII. Conclusion

This paper presents a bi-level optimization framework for crowd channel management that combines anisotropic geometric guidance with discrete directional constraints. The proposed SA-HBO algorithm exploits problem structure through surrogate pre-screening, Bellman warm-start, and block coordinate updates to handle the expensive simulation-based optimization efficiently.

The numerical experiments validate that: (1) geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local crowd behaviors; (2) the combined model exhibits stable sensitivity to configuration changes; (3) directional constraints effectively suppress counter-flow and head-on conflicts under bidirectional demand. These findings establish the foundation for subsequent strategy optimization studies.

Future work will extend to full multi-objective optimization, sensitivity analysis under demand perturbations, and calibration against real-world observation data.

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